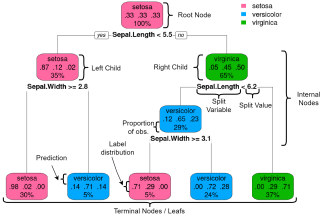
# Introduction to Machine Learning

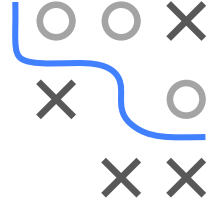
## Advanced Risk Minimization
## Loss functions and tree splitting




### Learning goals

- Know how tree splitting is 'nothing new' and related to loss functions
- Brier score minimization corresponds to gini splitting
- Bernoulli loss minimization corresponds to entropy splitting

# BERNOULLI LOSS MIN = ENTROPY SPLITTING



For an introduction on trees, see our **I2ML** lecture

When fitting a tree we minimize the risk within each node $\mathcal{N}$ by risk minimization and predict the optimal constant. Another common approach is to minimize the average node impurity $\text{Imp}(\mathcal{N})$.

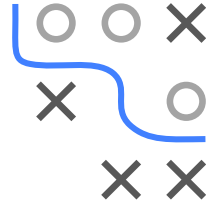
**Claim:** Entropy splitting $\text{Imp}(\mathcal{N}) = -\sum_{k=1}^{g} \pi_k^{(\mathcal{N})} \log \pi_k^{(\mathcal{N})}$ is equivalent to minimize risk measured by the Bernoulli loss.

Note that $\pi_k^{(\mathcal{N})} := \frac{1}{n_{\mathcal{N}}} \sum_{(\mathbf{x},y)\in\mathcal{N}} [y = k]$.

**Proof:** To prove this we show that the risk related to a subset of observations $\mathcal{N} \subseteq \mathcal{D}$ fulfills $\mathcal{R}(\mathcal{N}) = n_{\mathcal{N}}\text{Imp}(\mathcal{N})$,
where $\mathcal{R}(\mathcal{N})$ is calculated w.r.t. the (multiclass) Bernoulli loss

$$L\left(y, \pi(\mathbf{x})\right) = -\sum_{k=1}^{g} [y = k] \log\left(\pi_k(\mathbf{x})\right).$$

# BRIER SCORE MINIMIZATION = GINI SPLITTING



When fitting a tree we minimize the risk within each node $\mathcal{N}$ by risk minimization and predict the optimal constant. Another approach that is common in literature is to minimize the average node impurity $\text{Imp}(\mathcal{N})$.

**Claim:** Gini splitting $\text{Imp}(\mathcal{N}) = \sum_{k=1}^{g} \pi_k^{(\mathcal{N})} \left(1 - \pi_k^{(\mathcal{N})}\right)$ is equivalent to the Brier score minimization.

Note that $\pi_k^{(\mathcal{N})} := \frac{1}{n_{\mathcal{N}}} \sum_{(\mathbf{x},y)\in\mathcal{N}} [y = k]$

**Proof:** We show that the risk related to a subset of observations $\mathcal{N} \subseteq \mathcal{D}$ fulfills

$$\mathcal{R}(\mathcal{N}) = n_{\mathcal{N}} \text{Imp}(\mathcal{N}),$$

where Imp is the Gini impurity and $\mathcal{R}(\mathcal{N})$ is calculated w.r.t. the (multiclass) Brier score

$$L\left(y, \pi(\mathbf{x})\right) = \sum_{k=1}^{g} \left([y = k] - \pi_k(\mathbf{x})\right)^2.$$